Semester VII													
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
24INT451	Summer Internship	0	0	0	0	2	--	--	--	--	--	100
	Open Elective - 4 (From Other Department of FoET)	3	0	0	3	3	25	25	50	--	--	100
24PE431T	Flow Assurance	2	0	0	2	2	25	25	50	--	--	100
24PE432T	Corrosion Studies in Petroleum Industry											
24PE433T	Nanotechnology in Oil and Gas industry	3	0	0	3	3	25	25	50	--	--	100
25PE431T	City Gas Distribution											
24PE401T	Well Stimulation and Artificial Lift techniques	3	1	0	4	4	25	25	50	--	--	100
24PE402T	Integrated Reservoir Management and Enhanced Oil Recovery Methods	3	1	0	4	4	25	25	50	--	--	100
24PE403P	Reservoir Engineering Software Laboratory	0	0	2	2	1	--	--	--	50	50	100
24PE404T	Hydrocarbon Contracts and Asset Management	2	0	0	2	2	25	25	50	--	--	100
24PRPE451	Seminar	0	0	0	0	1	--	--	--	--	--	100
	Total	**16**	**2**	**2**	**20**	**22**						

Semester VIII												
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
24PRPE452	Major Project	0	0	0	0	12	-	-	-	-	-	100
24PRPE453	Comprehensive Project											
	Total					**12**						

24INT451					Summer Internship					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	0	2	0	-	-	-	50	50	100

COURSE OBJECTIVES

- The objective of the course is to expose students towards specialised aspect of chosen Oil and Gas Industry working practices.

INDUSTRIAL INTERNSHIP

The students are required to perform one month internship in the specialised aspect of chosen oil and Gas Industry

a) Acquaintance with the best working practices adopted in industry
b) To understand Industry working environment

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Develop Technical and Professional skills for employability in Oil and Gas industry
CO2- Develop professional communication skills (both written and oral).
CO3- Acquire understanding of project planning management methods and techniques.
CO4- Understand use of various artificial lift technique for various reservoir conditions
CO5- Demonstrate capability of team leadership and Team building for solving industry problems.
CO6- Develop skill sets for professional and ethical petroleum engineering practices.

24PE431T					Flow Assurance					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	-	-	100

COURSE OBJECTIVES

- To understand the concepts of flow assurance.
- To illustrate the analytical tools for flow assurance.
- To enhance skills to analyse and interpret basic descriptive and inferential statistics.
- To develop skills to address flow constraints in oil fields.

Unit I: Basics of Flow Assurance and Related Issues **Hours: 06**

Typical production system and flow assurance, Criteria of flow assurance, Resistances to the flow, various issues encountered for flow assurance in the industry, Importance of velocity profiles, temperature profiles and concentration profiles in prediction flow assurance issues, Importance of pressure drop calculations in flow assurance industry.

Unit II: Wax Deposition and Asphaltene Precipitation **Hours: 07**

Wax deposition definition, Wax deposition criteria, Mechanism of wax deposition, Factors for wax deposition, Wax deposition envelop, Models used to predict wax deposition, Definition of Asphaltenes, Criteria for formation of asphaltenes, Mechanism of asphaltenes precipitation, Factors responsible for asphaltenes precipitation, Asphaltenes precipitation envelop, Models utilized for prediction of asphaltenes.

Unit III: Inorganic Scale Deposition And Gas Hydrates Formation **Hours: 07**

Scale deposition definition, Scale deposition criteria, Scaling mechanism, Factors controlling scale depositions, Models used to predict scale deposition, Definition of gas hydrates, Criteria for formation of gas hydrates, Phase envelop for gas hydrate formation prediction, Models utilized for prediction of gas hydrates.

Unit IV: Mitigation of Flow Assurance and Case Studies **Hours: 06**

Need for flow assurance problem mitigation, Remedies and mitigation techniques for wax deposition, asphaltenes precipitation, scale deposition, and gas hydrates formation, Case studies on flow assurance in industry.

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Recognize a flow assurance problem and related issues.
CO2- Illustrate the need for flow assurance problem mitigation.
CO3- Apply knowledge to understand problems in details for wax deposition and asphaltenes precipitation.
CO4- Analyse the problem to know the root cause of scale deposition and gas hydrate formation.
CO5- Plan a remedial activity unique to the flow assurance problem mitigation
CO6-Understanding the flow assurance problems with case studies

TEXT / REFERENCE BOOKS

1. Shoham, O. (2006). *Mechanistic modeling of gas-liquid two-phase flow in pipes*. Society of Petroleum Engineers.
2. Frenier, W. W., Ziauddin, M., & Venkatesan, R. (2010). *Organic deposits in oil and gas production*. Society of Petroleum Engineers.
3. Frenier, W. W., & Ziauddin, M. (2008). *Formation, removal, and inhibition of inorganic scale in the oilfield environment*. Society of Petroleum Engineers.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE432T					Corrosion Studies in Petroleum Industry					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	-	-	100

COURSE OBJECTIVES

- To acquire the basic concept of corrosion mechanism and forms of corrosion
- To acquire the concept of corrosion testing
- To acquire the concept of the modern theory of corrosion
- To understand the basic mechanism and procedure of corrosion testing and prevention

Unit I: Fundamentals of Corrosion **Hours: 06**

Corrosion fundamentals, Corrosion in oil Industry, Cost of corrosion in the industry, Corrosivity of hydrocarbon fluids:- Water-oil, emulsion and multiphase flow regime, Wettability of metal surface, Corrosivity of aqueous phase in hydrocarbon fluids; Sulphur, and H_2S in hydrocarbon fluids; Influence of oil chemistry on the Corrosivity of the aqueous phase. Pipeline corrosion; Kinetics of electrochemical surface reactions; Cathodic reduction reactions; Anodic dissolution reactions; Transport of species; Transport from the bulk solution to the steel surface; Transport through the porous surface scales. Corrosion products; Kinetics of corrosion products precipitation and corrosion products growth.

Unit II: Forms of Corrosion **Hours: 07**

Modes of internal corrosion attack: Uniform corrosion; Localized corrosion ; Pitting corrosion; Erosion corrosion; Galvanic corrosion; Intergranular corrosion; Stress corrosion cracking; Hydrogen damage; hydrogen embrittlement; Hydrogen-induced cracking; Formation of hydride. Pipeline flow Corrosivity: Effect of water wetting; Effect of multiphase flow regime; Effect of multiphase velocity ; Effect of water phase characteristics; Significance of salinity; Significance of CO2 pressure; Significance of H_2S; Significance of O_2; Significance of pH; Effect of temperature. Materials selection:- Significance of alloying composition; Significance of steel microstructure.

Unit III: Testing of Corrosion **Hours: 07**

Experimental setups, methods, and standards:- Multiphase flow loop; Autoclave; Horizontal rotating cylinder; High velocity rig; Glass cell; Goniometer/Tensiometer ; Moisture content measurements ; Slow strain rate test. Corrosivity and corrosion rate determination:- Weight loss measurements; Potentiodynamic polarization and polarization resistance; Electrochemical impedance spectroscopy; Potentiostatic polarization

Unit IV: Pipeline Corrosion and Its Prevention **Hours: 06**

Pipeline Corrosion control; Environment control; Gas-phase contaminants and degasification; Water presence and dehydration/dewatering; Pipe cleaning; Pigging; Internal coating/liner; Chemical treatment and corrosion inhibitors:-Corrosion control by industrial inhibitors, Application methods; Influence of operating conditions; Solubility, partitioning, and compatibility. Biocides

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to
CO1- Understand the basic mechanism of corrosion process
CO2- Classify corrosion into different types and forms
CO3- Illustrate the influencing factors of pipeline corrosion
CO4- Demonstrate the procedure of corrosion testing
CO5-Explain the modern theory of corrosion based on thermodynamic and kinetics study
CO6-Interpret the preventive measures of corrosion in oil and gas industry.

TEXT / REFERENCE BOOKS

1. Papavinasam, S. (2014). *Corrosion control in the oil and gas industry*. Elsevier.
2. Cicek, V. (2015). Corrosion in the petroleum industry. In *Cathodic protection: Industrial solutions for protecting against corrosion* (pp. 231–245). John Wiley & Sons.
3. Nathan, C. C. (Ed.). (1973). *Corrosion inhibitors*. National Association of Corrosion Engineers (NACE).

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE433T					Nanotechnology in Oil and Gas industry					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To develop skills to address flow constraints in oil fields. Understand the principles of nanotechnology and nanomaterials relevant to the petroleum industry with the synthesis, characterization, and physicochemical properties of nanomaterials.
- Analyze the applications of nanotechnology in drilling, cementing, reservoir engineering, enhanced oil recovery, production, and environmental management.
- Explore recent advances and future trends in nanotechnology for the energy sector.

Unit I: Fundamentals of Nanotechnology and Nanomaterials **Hours: 10**

Introduction to nanotechnology and its evolution; Classification of nanomaterials; Size-dependent physical and chemical properties; Surface area, surface energy, and quantum effects; Types of nanoparticles: metal, non-metal oxide, carbon-based, polymeric, and nanocomposites; Synthesis methods: Top-down and Bottom-up approaches; Green synthesis of nanoparticles; Characterization techniques: SEM, TEM, XRD, FTIR, DLS, Zeta potential, BET, TGA; Stability and dispersion of nanoparticles; Health, safety, and environmental considerations

Unit II: Nanotechnology Applications in Drilling, Cementing, and Reservoir **Hours: 10**

Nanoparticles in water- and oil-based drilling fluids; Rheology modification and fluid-loss control; Wellbore stability and shale inhibition; High-pressure high-temperature (HPHT) drilling applications; Nanotechnology in oil-well cementing; Different nanoparticles in cement properties modification; Self-healing and smart cement systems; Nanotechnology for reservoir characterization; Nano-sensors and pore-scale imaging; Formation damage diagnosis and mitigation

Unit III: Nanotechnology for Production and Flow Assurance **Hours: 10**

Fundamentals of Nano-Enhanced Oil Recovery (Nano-EOR); Nanoparticles transport in porous media; Colloidal stability and adsorption; Wettability alteration mechanisms; Interfacial tension reduction; Structural disjoining pressure; Mobility control; Nano-surfactant flooding; Nano-polymer flooding; Hybrid chemical EOR processes; Wax, asphaltene, and scale control; Corrosion inhibition using nanoparticles; Gas hydrate mitigation

Unit IV: Emerging Trends in the Hydrocarbon Industry with case studies **Hours: 09**

Nanotechnology in production optimization; Produced water treatment; Oil spill remediation and wastewater treatment; Nanotechnology in carbon capture, utilization, and storage (CCUS); Nanotechnology for hydrogen and geothermal energy; AI-enabled smart nanomaterials and digital oilfields; Industrial case studies and future trends in nanotechnology

Max <39Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1-Explain the fundamental principles, classification, and properties of nanomaterials used in the oil and gas industry.
CO2-Describe the synthesis and characterization techniques employed for petroleum-related nanomaterials.
CO3-Analyze the applications of nanotechnology in drilling, cementing, and reservoir characterization.
CO4-Evaluate the mechanisms and performance of nanotechnology-assisted enhanced oil recovery (Nano-EOR) processes.
CO5-Assess the role of nanotechnology in production engineering, flow assurance, and environmental sustainability.
CO6-Recommend suitable nanotechnology-based solutions to address contemporary challenges in the oil and gas industry.

TEXT / REFERENCE BOOKS

1. Shah, M. A., & Ahmad, T. (2010). *Principles of nanoscience and nanotechnology*. Alpha Science International.
2. Suleimanov, B. A., Veliyev, E. F., & Vishnyakov, V. (2022). *Nanocolloids for petroleum engineering: Fundamentals and practices*. Wiley.
3. Huh, C., Daigle, H., Prigiobbe, V., & Prodanović, M. (2019). *Practical nanotechnology for petroleum engineers*. CRC Press.
4. Taborda, E., Villada, Y., Giraldo, L. J., Estenoz, D. A., Franco, C. A., & Cortés, F. B. (2024). *Application of nanotechnology in the petroleum industry: A view from rheology*. IntechOpen.
5. Gangawane, K. M., & Kumar, A. (2025). *Nanotechnology in enhanced oil recovery*. CRC Press.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

25PE431T					City Gas Distribution					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- Identify the various attributes of CGD value chain.
- Understanding the policies pertaining to CGD as prescribed by PNGRB.
- Examine the various engineering aspects pertaining to the CGD network.
- Analysing the components of CNG and PNG sector.
- Assessing the responsibilities for HSE in CGD industry.
- Simulating the optimisation of CGD network using IT.

Unit I: Introduction of Natural Gas and Its Value Chain **Hours: 10**

Types of Natural Gas, Introduction to concept of Geographical Area (GA), History of CGD in India; LNG, LPG, and CGD business. Gas Value Chain**:** Gas Transmission and Distribution System; City Gate Station (CGS); Gas Filtration and Pressure reduction skids; Odorizing unit; Common pressure reduction station (CPRS)/District Regulation Station (DRS); Meters including smart meters; Pipeline for CGD; Steel and PE Pipelines; CNG infrastructure: Mother Station, Online Station, Daughter Station, Daughter Booster Station;

Unit II: Petroleum and Natural Gas Regulatory Bodies **Hours: 10**

Introduction to Ministry of Petroleum and Natural Gas (MoPNG); Upstream and downstream regulators, Powers and functions of Petroleum and Natural Gas Regulatory Board (PNGRB), Evolution of CGD bidding in India till date, Bidding attributes and Bid authorization, Natural gas trend, Technical Standards and Specifications including Safety Standards (T4S).

Unit III: Accessory Equipment's and Its Applications in CGD Value Chain **Hours: 10**

Annual O&M Plan; Valves, Pumps, Compressors, Welding, Cathodic Protection System, QHSE: CNG Safety; Emergency Response Plan; Disaster Management Plan; Quality assurance concepts; Inspection and Surveillance; Risk Assessment in CGD Business; Fire and Fire Protection equipment'

Unit IV: Advance Methods in CGD Value Chain **Hours: 09**

Types of Cylinders, HCNG, LCNG, Compressed Biogas, Dispenser, IT in CGD including Artificial Intelligence (AI), Machine Learning (ML), Internet of Things (IoT), Block chain and its application in centre.

Max <39Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Summarize City Gas Distribution value chain and Natural Gas system.
CO2- Acquaint and adapt the roles, functions and objectives of PNGRB
CO3- Explain Piped Natural Gas (PNG) distribution system.
CO4- Classify the types of Compressed Natural Gas (CNG) Stations and explain the CNG infrastructure.
CO5- Practice the HSE measures for safety of CGD sector.
CO6-Assess the steps to be taken in financing the CGD measures

TEXT / REFERENCE BOOKS

1. Sircar, A., & Yadav, K. (2020). *Natural gas: A comprehensive study*. Technology Publishers.
2. Sircar, A., Sahajpal, S., & Modi, U. (2017). *City gas distribution: An Indian perspective*. Technology Publishers.
3. Infraline Energy. (2018). *City gas distribution in India: Demystifying the opportunity, growth and investment potential*. Infraline Energy.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE401T					Well Stimulation and Artificial Lift Techniques					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	-	-	100

COURSE OBJECTIVES

- Demonstrate the concepts of Production enhancement
- Develop Skills to design Stimulation Jobs
- Demonstrate the use of artificial lift techniques.
- Develop skill to analyse the best suitable artificial lift solutions for low productivity wells.

Unit I: Workover and Well Stimulation Techniques **Hours: 18**

Introduction to well stimulation; Well analysis and remedial measures, Principles and application of workover methods, workover rig, Coil tubing unit, workover fluids, Water Control Techniques-Reasons, identification and control techniques, Sand Control Techniques Reasons, identification and control techniques. Well stimulation methods to mitigate formation damage problems; Acid wash; Matrix stimulation; Design of matrix acidization; Acid fracturing. Basics of hydraulic fracturing and its applications: Candidate well selection; Effect of Rock mechanical properties; Stress profile and fracture height containment, well placement; Proppants characteristics and its selection; Fracturing fluid systems and additives; Fracture pressure analysis and step-rate and leak off tests; Treatment stage design, Calculations of clean and slurry volumes for treatment designs, surface pumping pressure and no. of pumps and other relevant topics.

Unit II: Multi-Stage Hydraulic Fracturing and Re-Fracturing **Hours: 10**

Well-pad configuration; Well spacing and Fracture Driven Interactions; Multi-stage fracturing: Stage length and cluster spacing, Multistage fracturing techniques: Plug & perf, sliding sleeves, Fluid diverters; Proppants selection; Refracturing: benefits and limitations, candidate selection, procedures; Diagnostic methods: microseismic, fiber sensing-DAS and DTS, radioactive tracers, and tiltmeter. Modelling 2D, Pseudo3D, and 3D models; Flow chart of fracture simulator. Case studies on multi-stage fracturing and refracturing treatments; advanced fracturing techniques.

Unit III: Artificial Lift Technologies **Hours: 07**

Introduction to Artificial Lift Techniques: Artificial lift methods, Rod Pumping: Reciprocating rod lift, progressive cavity pump, Electric submersible Pump (ESP), Hydraulic lifts (Jet Pump), Gas Lift: Continuous, Intermittent, Plunger assisted lift.

Unit IV: Design, Monitoring and Optimization **Hours: 17**

Sucker Rod Pump, Electric submersible pump, progressive cavity pump, Gas lift: working principles, Surface and Subsurface components, Designing, Monitoring and Optimization, Common Operating Problems and its mitigation. Digital technologies and latest developments in the artificial lift industry.

Max <52Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Understand various well stimulation techniques and its application for various reservoir conditions
CO2- Design of hydraulic fracturing treatment stages, proppant type and fracturing fluids
CO3- Select the candidate and prepare the workflow for designing multi-stage fracturing and refracturing treatments
CO4- Understand use of various artificial lift technique for various reservoir conditions
CO5- Understand the working principle of various artificial lift used in the oil and gas industry
CO6- Design of artificial lift technique like sucker rod pump, gas lift, electric submersible pump, progressive cavity pump

TEXT / REFERENCE BOOKS

1. Guo, B. (2007). *Computer-aided petroleum production engineering*. Gulf Publishing Company.
2. Miskimins, J. L. (2019). *Hydraulic fracturing: Fundamentals and advancements*. Society of Petroleum Engineers.
3. Economides, M. J., & Nolte, K. G. (2000). *Reservoir stimulation* (3rd ed.). John Wiley & Sons.
4. Brown, K. E. (1984). *The technology of artificial lift methods* (Vols. 1–4). PennWell Corporation.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE402T					Integrated Reservoir Management and Enhanced Oil Recovery Methods					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	-	-	100

COURSE OBJECTIVES

- To demonstrate the concepts of Reservoir management in oil and gas exploitation.
- To imbibe the knowledge of reservoir engineering in E & P business.
- To develop Skills to analyse reservoir behaviour and monitoring plan for reservoir management.
- To develop skills to implement best reservoir optimization practices for enhanced recovery.

Unit I: Concept Reservoir Management & Characterization **Hours: 12**

Concept of integrated reservoir management, Evolution of reservoir management with time, Fundamentals of reservoir management, synergic team approach; Importance of reservoir management, Reservoir Data types: Geosciences, Fluid and PVT, Petrophysical properties from core, Well logs, Transient well test, and Production, Data validation and workflow development for reservoir characterization.

Unit II: Reservoir Performance Analysis & Development Plans for Techno-Economic Evaluation **Hours: 15**

Reservoir performance analysis by various methods: Volumetric, decline curve, material balance & simulation, Concept of field development plan, Importance of EOR/IOR in reservoir management strategy, Development plans for newly explored/developed and matured fields, Scenarios for development plans & Techno-economic evaluation. Risk management, Uncertainties & economic optimization.

Unit III: Enhanced Oil Recovery Methods **Hours: 15**

Definition of enhanced oil recovery and improved oil recovery, Difference of IOR and EOR, Target oil resource for EOR, General Classification. Description and potential of different EOR processes, Microscopic and macroscopic displacement of fluids in a reservoir, Displacement efficiency in different system – linear, areal, volumetric, Definition and discussion of mobility ratio and mobility control processes for different types of fluids, Miscible/Immiscible displacement processes - water flooding, gas injection, micro-emulsion flooding Chemical Flooding - polymer flooding, Surfactant flooding, Micellar flooding related methods Thermal recovery processes- in situ combustion, hot-water injection, steam flooding, SAGD Microbial EOR. EOR screening criteria.

Unit IV: Case Studies on Reservoir Management and Enhanced Oil Recovery **Hours: 10**

Field scale implementation and their performance of various EOR schemes of local and global context, Reservoir Management case studies for various types of clastic and non-clastic reservoirs (both onshore and offshore), Current challenges and areas of further development.

Max <52Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Understand the integrated reservoir management concept and synergistic team approach for reservoir management and characterization
CO2- Integrate the fluid, Petrophysics and production data of a reservoir for analysing its static and dynamic performance.
CO3- Understand the concept of field development in distinct stages in the life cycle of a reservoir with input from reservoir modelling and simulation
CO4- Analyse the key issues and able to prepare field development plan for favourable risked-techno-economic scenario.
CO5- Understand the enhanced oil recovery methods for additional oil recovery.
CO6- Design and execute the enhanced oil recovery methods and reservoir management plan.

TEXT / REFERENCE BOOKS

1. Satter, A., & Thakur, G. C. (1994). *Integrated petroleum reservoir management: A team approach*. PennWell Publishing Company.
2. Kumar, S. (1987). *Development of oil and gas fields*. Allied Printers.
3. Dake, L. P. (1978). *The practice of reservoir engineering*. Elsevier Scientific Publishing Company.
4. Green, D. W., & Willhite, G. P. (1998). *Enhanced oil recovery*. Society of Petroleum Engineers.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE403P					Reservoir Engineering Software Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

- To provide a deep understanding of various Professional/Industrial software tools used in the Reservoir Engineering Domain
- To provide the significance of Reservoir modelling and simulations for various Reservoir engineering problems.

LIST OF EXPERIMENTS

1. Seismic Data Loading, Visualization and Processing using Kingdom suite/ tNavigator – Part1.
2. Seismic Data Loading, Visualization and Processing using Kingdom suite/ tNavigator – Part2.
3. Petrophysical Investigations using Well Log Analysis.
4. Development and validation of Geo-cellular Modelling using Geostatic prediction and Variogram Modelling – Part1.
5. Development and validation of Geo-cellular Modelling using Geostatic prediction and Variogram Modelling – Part2.
6. Fluid Property (PVT) Modelling for Black Oil or Compositional Oil Simulations.
7. Simulation of SPE Comparative Solution - Model 1 (SPE-9723-PA) – Part1.
8. Simulation of SPE Comparative Solution - Model 1 (SPE-9723-PA) – Part2.
9. Assisted History Matching using an Optimization Algorithm – Part1.
10. Assisted History Matching using an Optimization Algorithm – Part2.
11. Simulation of Enhanced Oil recovery Process
12. Well Testing Studies using Kappa Engineering
13. Fracture design, analysis and optimization using FracPRO – Part1.
14. Fracture design, analysis and optimization using FracPRO – Part2.

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 -Demonstrate the role of various software tools in reservoir engineering

CO2 -Classify the various software tools available in the Reservoir domain for history matching Performance prediction and Risk analysis.

CO3- Create a Static and Dynamic reservoir simulation model using simulation software's.

CO4 -Excel the fundamental modelling workflows associated with the simulation software.

CO5 -Comprehend complex and dynamic nature of the Reservoir engineering problems including Pressure transient analysis, hydro-fracturing etc. and formulate a solution strategy for effective management at the field scale.

CO6-Identify the best tool matching the type and scope of the numerical study deployed to perform in the future.

TEXT / REFERENCE BOOKS

1. Software manuals
2. Odeh, A. S. (1981). Comparison of solutions to a three-dimensional black-oil reservoir simulation problem. *Journal of Petroleum Technology, 33*(1), 13–25.

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

Evaluation based on the class performance and Laboratory book **50 Marks**

End semester examination and Viva-voce **50 Marks**

24PE404T					Hydrocarbon Contracts and Asset Management					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	-	-	100

COURSE OBJECTIVES

- Demonstrate legal principles of oil and gas contracts
- Illustrate Marketing strategies in Oil industry
- Imbibe Portfolio Management skills in graduates
- Develop skills to address dispute in oil business

Unit I: **Hours: 06**

Life cycle of Petroleum Project, Fiscal System in hydrocarbon industry (Progressive, regressive fiscal policy tax and non-tax components of Fiscal Policy), Basic elements of Contracts, Basic terminologies of contract and legal. Basics of Upstream and Downstream regulatory Laws and Policies, Fundamentals of Oil and gas accounting: Capex, Opex, Cost classification, Depreciation depletion and amortization (DDA), Cash flow, internal rate of return, Net Present value

Unit II: **Hours: 07**

Upstream Agreements (Concessionary, Production sharing, Risk sharing,), Indian policy on Hydrocarbon Exploration and Licensing Policy (HELP). Drilling Contracts, Farm-in and Farm-out, Joint ventures, Comingling allocation and attribution agreement, Gas sale and supply agreement. LNG Agreement

Unit III: **Hours: 06**

Hydrocarbon trading-Oil trading, Physical and Paper; Crude oil Markets- Spot, Barter, Future and forward. Oil pricing mechanism, short term and long term Oil Pricing

Unit IV: **Hours: 07**

Asset Integrity Management Introduction to concept of Asset Management& Asset Integrity Management, The Asset Management System – Asset Management Policy – Asset Management Strategy – (Strategic) Asset Management Plans – line of sight; International standard on Asset Management: ISO 55000; Risk & Risk Assessment Approaches Used for Asset Integrity Management; Identification & assessment of risk; Risk management: using the risk matrix, risk register & hazard log; Risk Based Maintenance Deterioration: the way assets could fail the seven steps of Risk Based Maintenance (RBM) / Reliability Centred Maintenance (RCM) incl. Failure Mode Effects &Criticality Analysis; Failure behaviour of onshore & offshore systems.

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Master and comprehensively understand the legal principles of oil and gas contracts
CO2- Understanding in depth licensing, production sharing and service contracts
CO3- Identifying the risk factors and managing those risks through effective contractual clauses
CO4- Know the relevant legal and regulatory frameworks that are in the oil and gas industry
CO5- Learn how to use appropriate contractual clauses in oil and gas contracts
CO6-Appreciate the best dispute resolution methods and how it will apply it in different scenarios.

TEXT / REFERENCE BOOKS

1. Shippey, K. C. (2009). *A short course on international contracts* (4th ed.). World Trade Press.
2. Tordo, S. (2007). *Fiscal systems for hydrocarbons: Design issues*. The World Bank.
3. Ministry of Petroleum and Natural Gas. (2016). *Hydrocarbon Exploration and Licensing Policy (HELP) and model production sharing contract*. Government of India.
4. Johnston, D. (1994). *International petroleum fiscal systems and production sharing contracts*. PennWell Books.
5. Wright, C. J., & Gallun, R. A. (2008). *Fundamentals of oil and gas accounting* (5th ed.). PennWell Books.
6. Millar, M. P. (2015). *Asset integrity management handbook*. Gulf Professional Publishing.
7. Jennings, A. (2002). *Oil and gas exploration contracts*. PennWell Books.
8. Davidson, M. R. (1999). *Oil and gas infrastructure and midstream agreements*. Cameron May.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

PART A: 10 Questions of 2 marks each-No choice **20 Marks**

PART B: Each unit carries 20 marks, with internal choice **80 Marks**